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Roll no.

Degree: B. Tech Semester: 3rd — Course work

MID-SEMESTER EXAMINATION, September, 2025

Course Title: **Probability Theory and Random Process**

Course Code: **EAEPC303**

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Two cards are drawn in succession from a deck of cards. Find the chance that the first is a king and the second is the queen if the first card is (a) Replaced (b) Not Replaced.	2	CO1
1b	A Fair coin is flipped 10 times. Find the probability of occurrence of 5 or 6 heads by considering suitable distribution of the random variable.	2	CO1
2a	Compute Mean square value and Variance of a Uniformly distributed random variable over an interval.	2	CO2
2b	The PMF of a discrete random variable X is given by, $P_X(x) = \begin{cases} \frac{1}{3}, & x = -2, 0, 2 \\ 0 & \text{otherwise} \end{cases}$ Find the mean and the variance of X .	2	CO2
3a	The probability density function of a random variable X is given by, $f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ Find the mean and the variance of X .	2	CO2
3b	State and define joint cumulative distribution function and explain the significant properties.	2	CO2
4a	If X and Y are independent random variables, show that they are uncorrelated	2	CO2
4b	Explain joint density function and marginal density function.	2	CO2
5a	Define ij -th central moment of two random variables, X and Y . Write down the expression of covariance of X and Y and explain its significance.	2	CO2
5b	If X and Y are uncorrelated then prove that $\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2,$ where $\sigma_{X+Y}^2 = E[(X+Y)^2] - (E[X+Y])^2$	2	CO2